

09 · RENDERS Y REPRESENTACIÓN · CONTROL GEOMÉTRICO

El 3D como base de control

Doce vistas hemos vuelto al blanco completo de control geométrico: beauty, neutro, clay con y sin oclusión, profundidad y normales en píxeles y más vistas de familia. La base sobre la que la fase de imágenes AI podrá ser atmosférica, vegetal y viva SIN redefinir el proyecto.

VISTAS	PASSES POR VISTA	RESOLUCIÓN	ESCENA
12	8	1920×1200	v006 build 12

RP-00 · PROPÓSITO Y REGLAS DE USO

La IA potencia; el 3D manda

— **beauty.png** imagen base con materiales y luz PSA de tarde: referencia de color, escala y relación urbana. Punto de partida img2img de denoise moderado.

— **depth.png + normal.png** control duro de geometría (tipo ControlNet depth/normalbae): la IA no puede mover un mástil ni engordar la cinta.

— **edges.png** control de línea (Sobel de depth+normal): conserva barandillas, costillas y perfiles finos que el depth solo no protege.

— **id_mat.png** máscaras por familia (granito/vegetación/rosa/agua/contexto/madera/canopy/monumento): inpainting selectivo — enriquecer la vegetación SIN tocar la estructura, revivir el agua SIN alterar fachadas.

— **clay.png / clay_sinveg.png** lectura volumétrica pura y variante sin verde: para regenerar la vegetación completa con IA manteniendo la estructura exacta.

— **ao.png** oclusión pura: refuerzo de contacto y sombra propia en el compositing.

— **CAMERA_METADATA.json** cámara exacta reproducible: posición, target, matriz mundo, FOV vertical, focal equivalente, clipping y resolución.

RP-01 · REGLA DE FIDELIDAD JERÁRQUICA

NIVEL	ELEMENTOS	QUÉ PUEDE HACER LA IA
HARD LOCK	arquitectura, cámara, estructura, pasarela, escaleras, ascensores, canopies, mástiles, pods, agua construida, edificios, Metro, Carlos III, geometría urbana principal	NADA: coincidencia con edges/depth; cualquier mutación descarta la imagen
SOFT LOCK	jardineras, mobiliario, distribución y masa general de la vegetación	variar densidad y textura DENTRO de su huella id_mat; no crear ni borrar piezas
CONTROLLED FREEDOM	hojas individuales, pequeñas ramas, personas, vapor, salpicaduras, reflejos, textura superficial, imperfecciones, atmósfera	libertad total: aquí vive lo fotográfico y lo vivo

edges.png impide el rediseño estructural; NO exige coincidencia píxel a píxel de fenómenos orgánicos. El piloto (una sola vista, H1) congela el método antes de aplicarlo al resto: pack › generación › comprobación geométrica › corrección › versión final.

PACK · H1

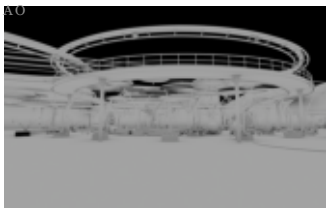
Hero general a nivel de ojo

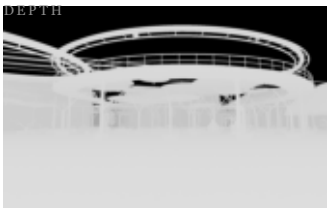
plaza completa: cinta, mástiles, dosel, gotas

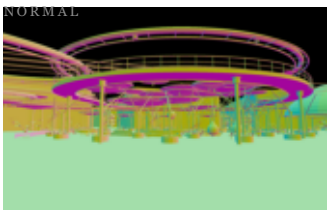


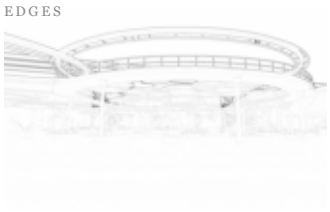


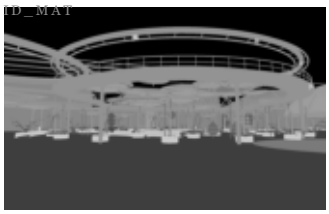












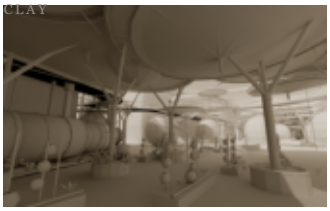
CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H1	-2.0 -18.0 1.6	6.0 16.0 5.0	62.0°	20 mm	0.10–1200 m

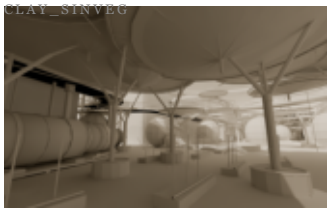
PACK · H2

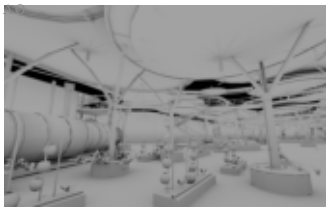
Frente oeste

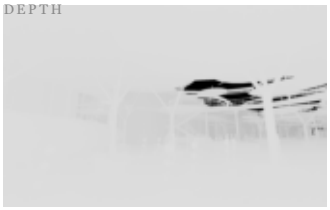
copas C1 con escalera-viga y jardineras de borde

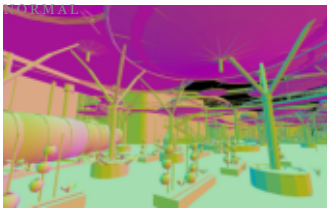


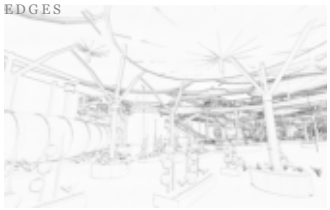


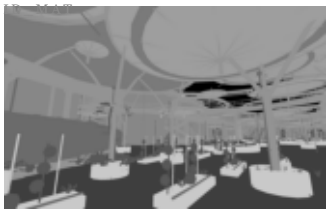










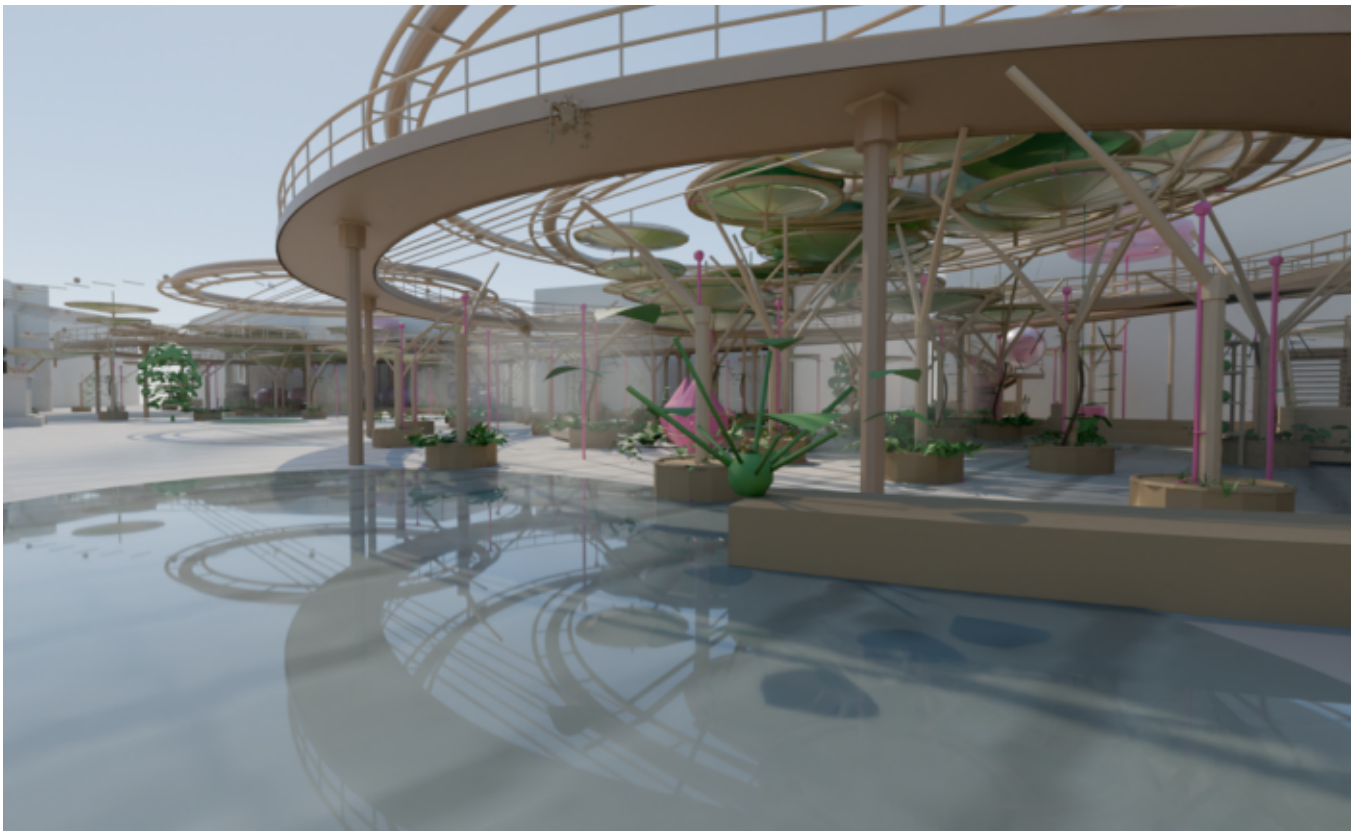


CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H2	-56.0 2.0 1.6	-44.0 18.0 5.5	68.0°	18 mm	0.10–1200 m

PACK · H3

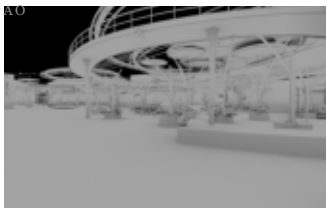
La lámina jugable

splash y estanques con la estructura detrás

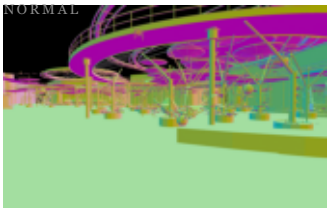




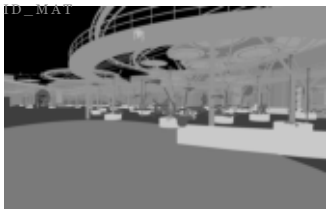












CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H3	24.0 -10.0 1.6	14.0 -1.0 1.2	66.0°	18 mm	0.10–1200 m

PACK · H4

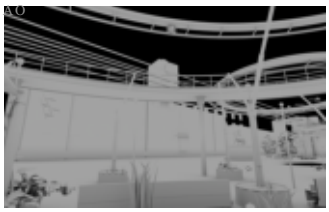
Corredor visual a Correos

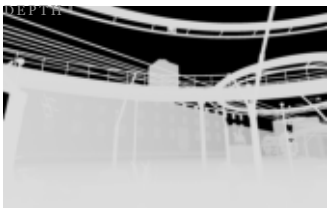
el dosel enmarcando torre y reloj

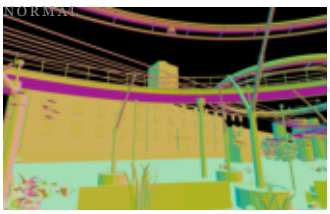


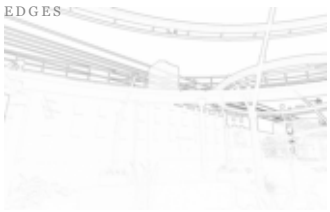


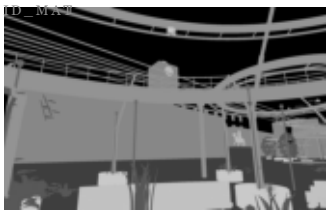










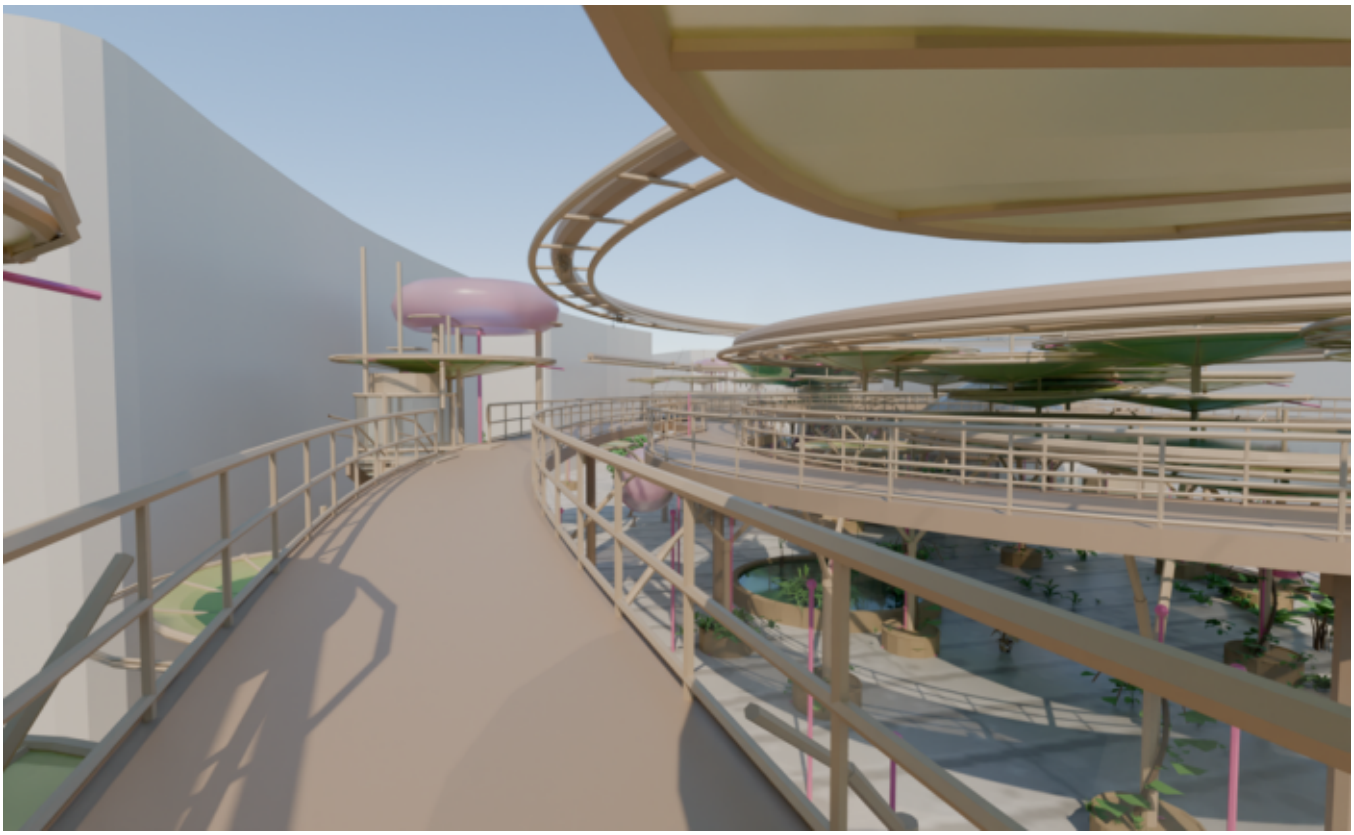


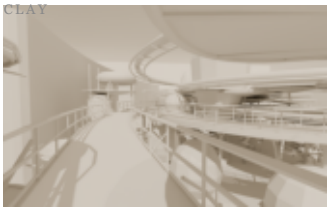
CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H4	6.0 8.0 1.6	-25.2 -36.5 15.0	58.0°	22 mm	0.10–1200 m

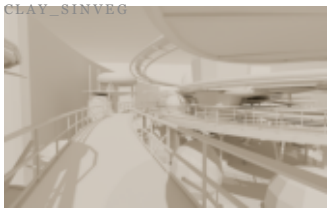
PACK · H5

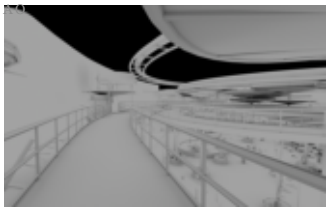
Sobre la cinta

el mirador en movimiento a cota +6

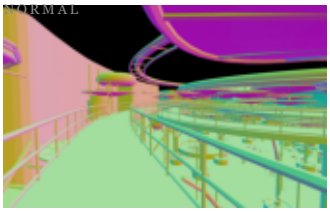


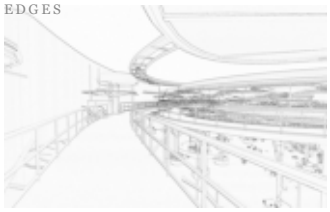


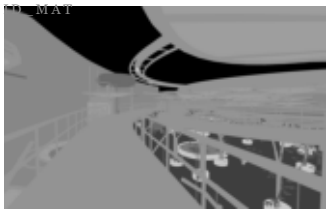










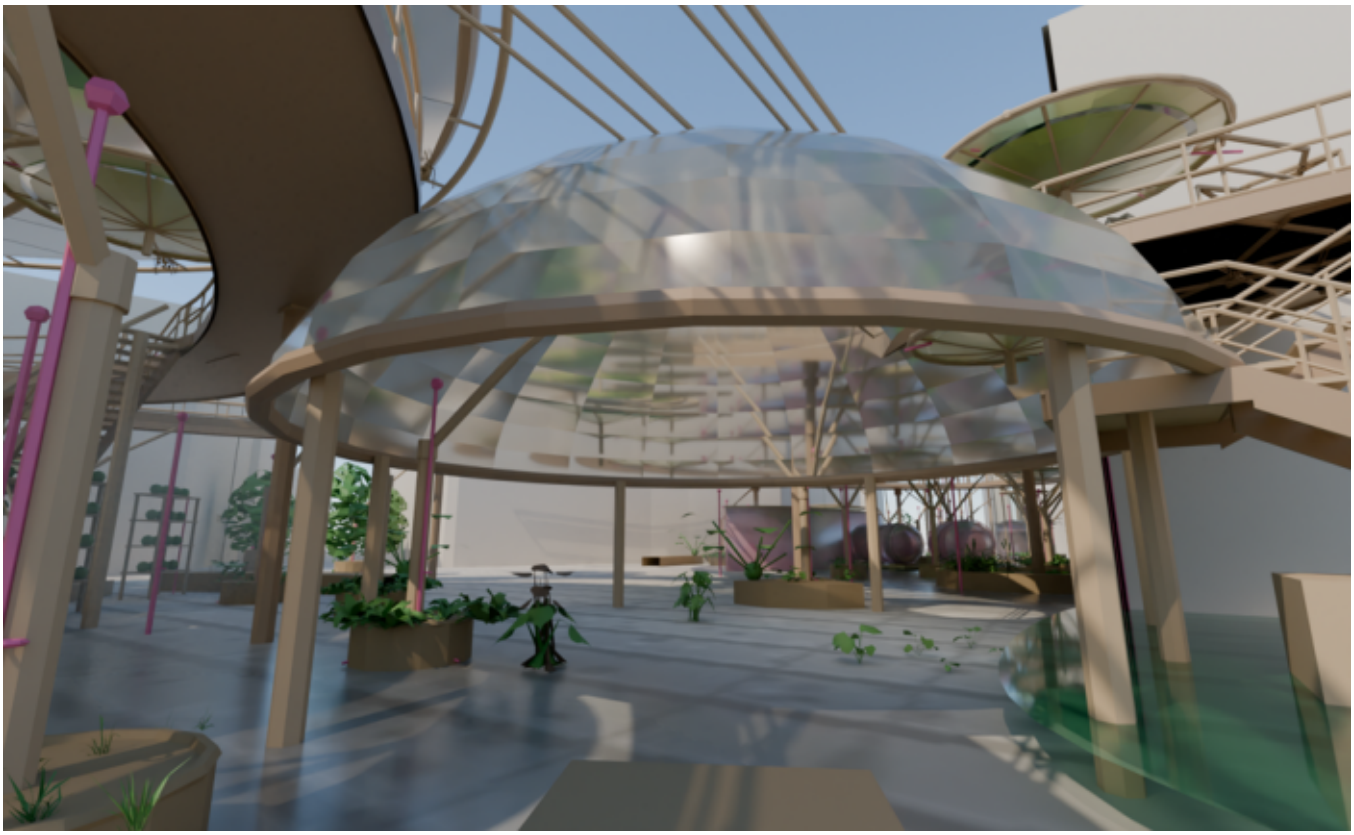


CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H5	-12.8 28.8 7.5	10.0 26.0 6.5	63.0°	20 mm	0.10–1200 m

PACK · H6

Entre mástiles

el bosque estructural con los pods rosa



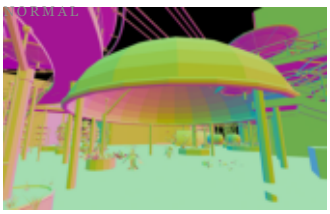




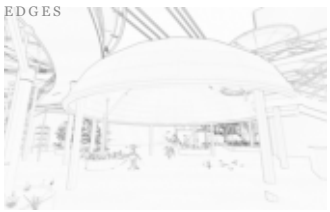


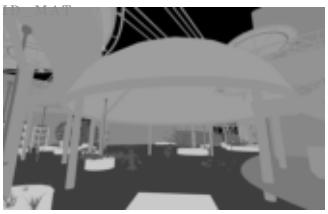
DEPTH





EDGES



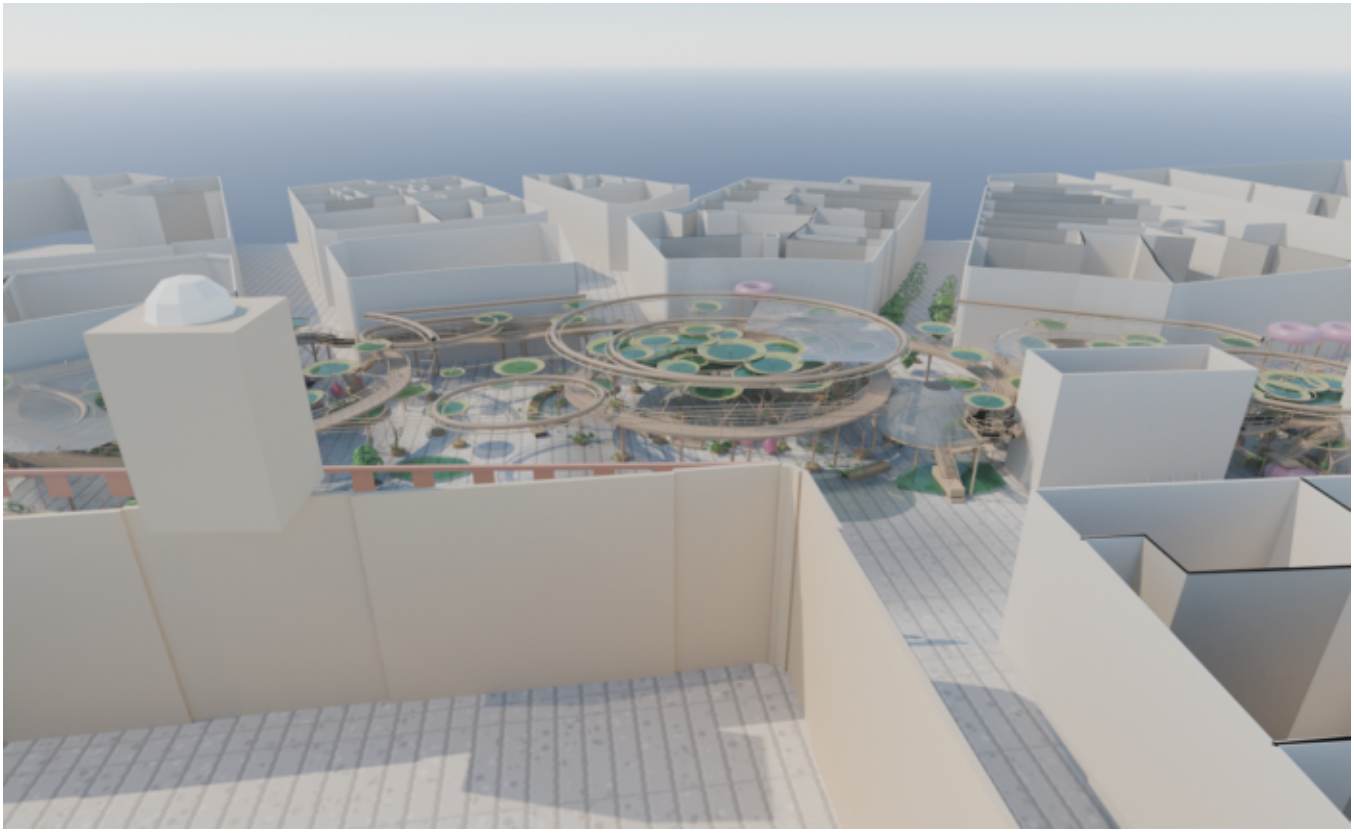


CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H6	22.0 -4.0 1.6	33.0 5.0 4.5	68.0°	18 mm	0.10–1200 m

PACK · H7

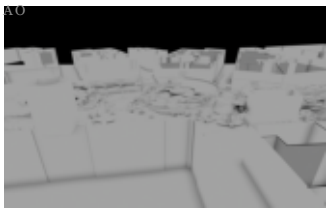
Aérea general

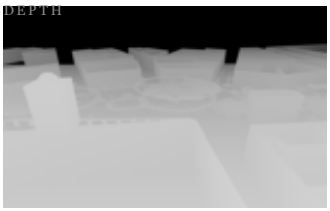
las tres copas, la cinta y el claro ceremonial

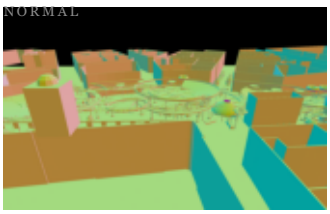


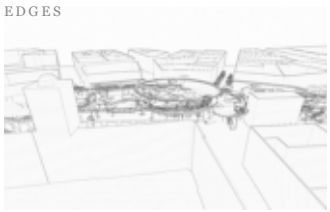


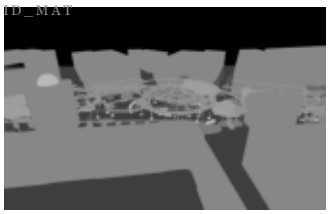










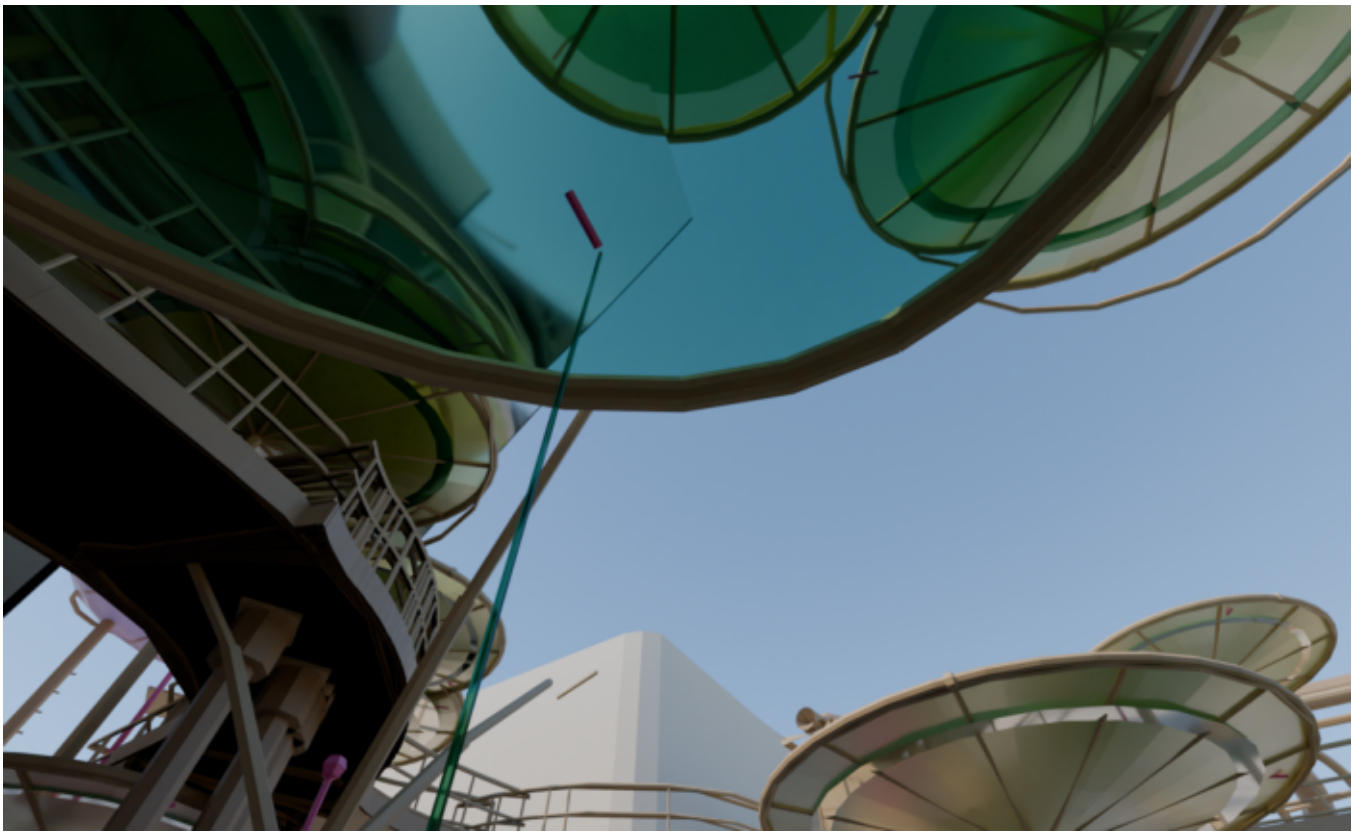


CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
H7	0.0 -80.0 42.0	2.0 12.0 0.0	55.0°	23 mm	0.10–1200 m

PACK · F1_INTERIOR_DOSEL

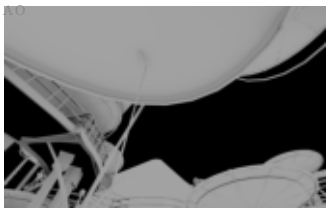
Interior del dosel

mirando arriba: hojas y microalgas

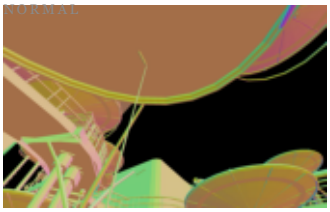


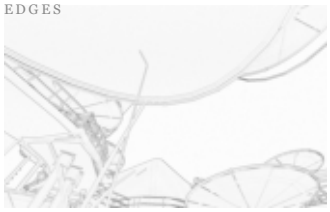


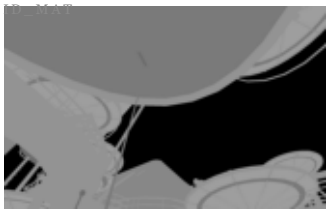










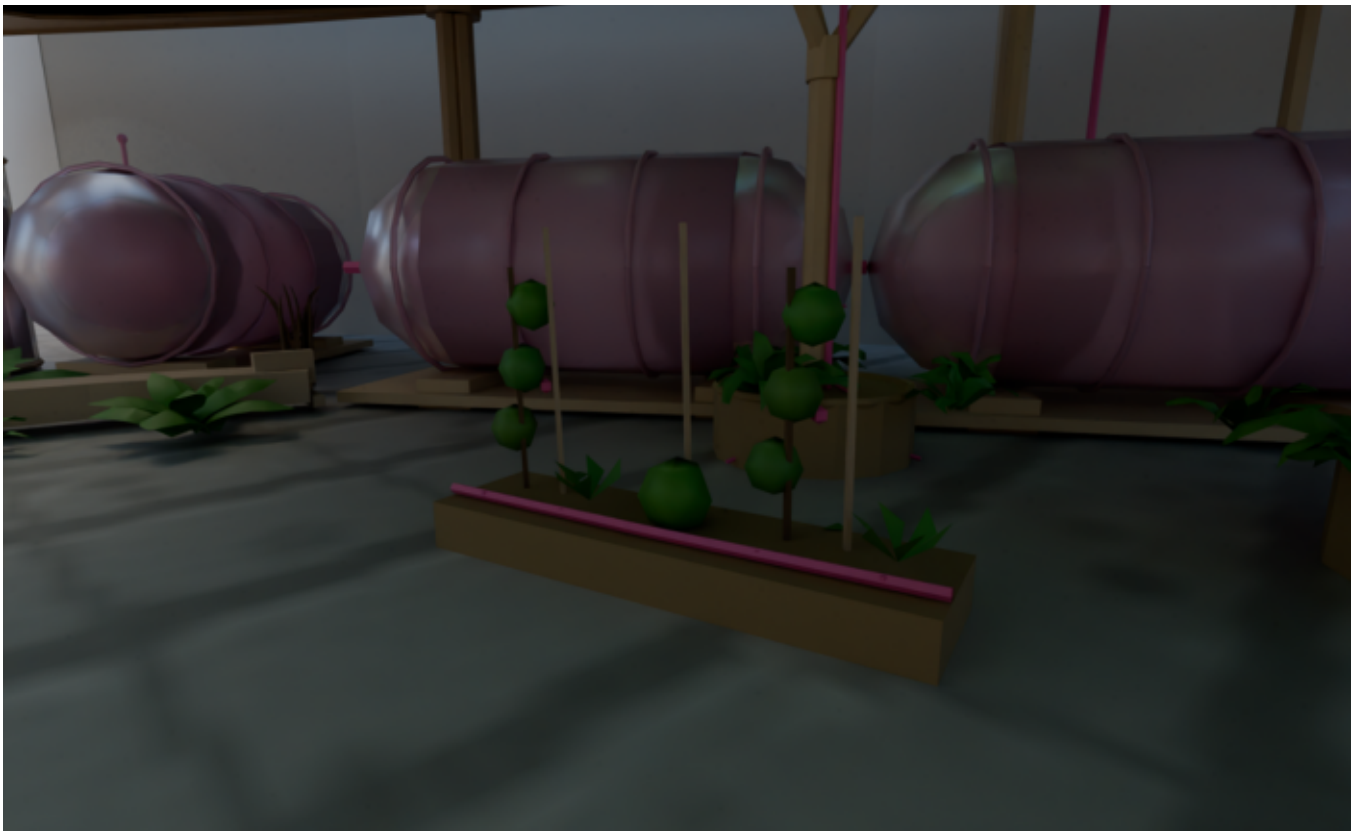


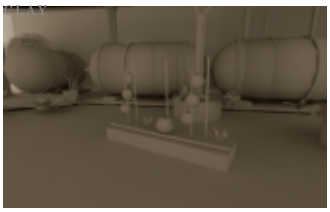
CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
F1_INTERIOR_DOSEL	-50.0 12.0 1.6	-47.0 17.0 10.0	70.0°	17 mm	0.10–1200 m

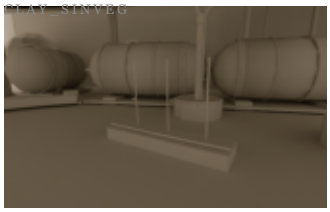
PACK · F2_HUERTO

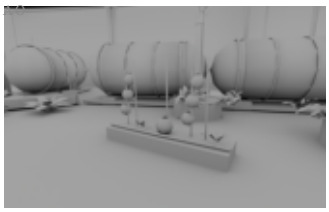
El huerto C1

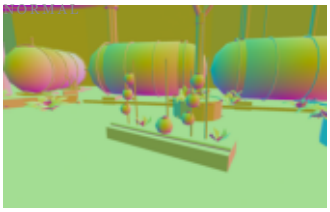
maíz por fases, goteo, mulch



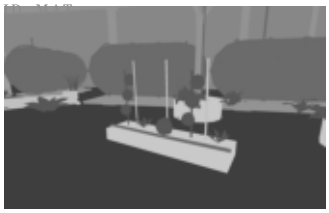










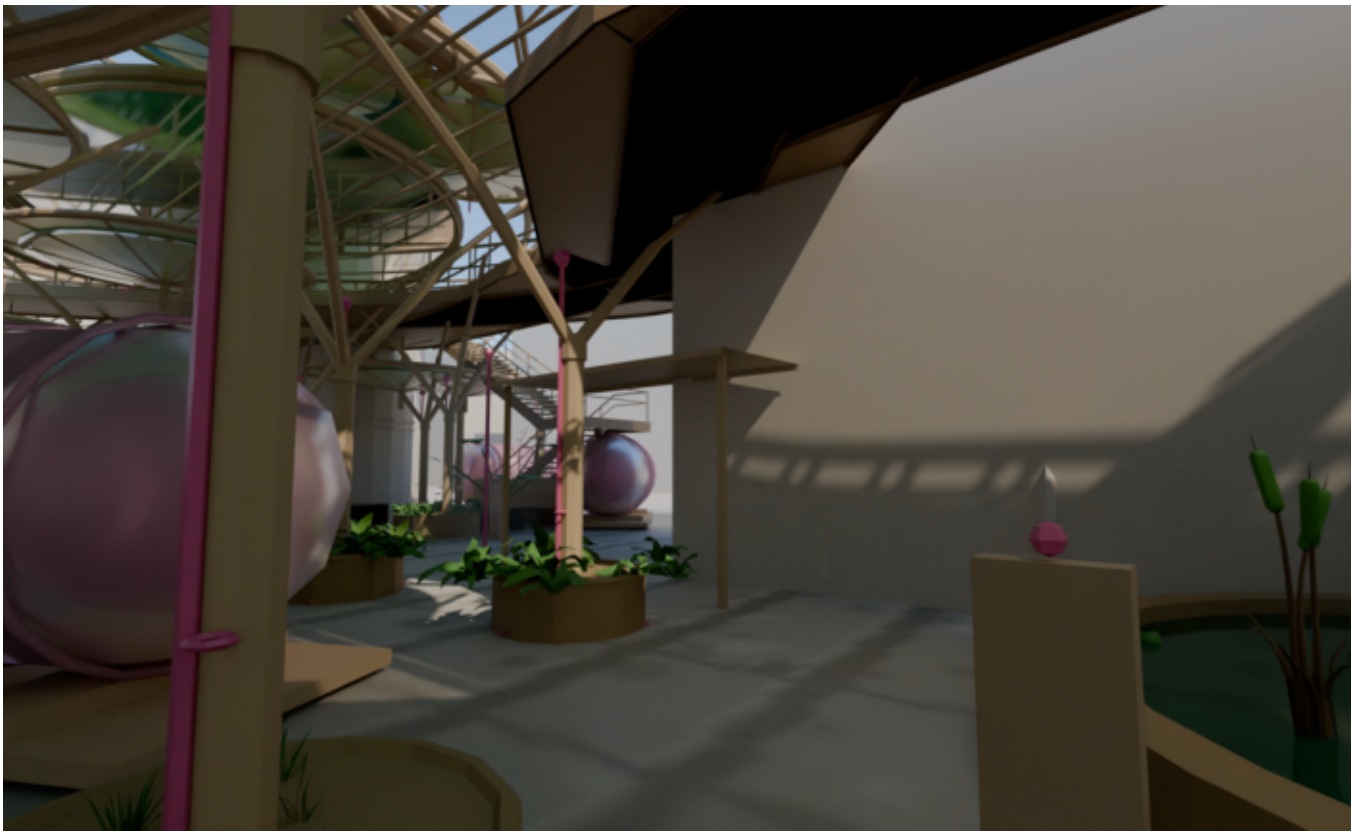


CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
F2_HUERTO	-54.0 4.0 1.5	-56.5 6.2 0.8	60.0°	21 mm	0.10–1200 m

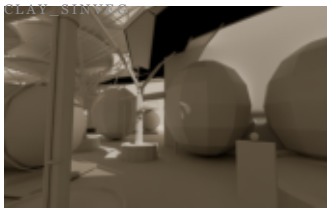
PACK · F3_ATRIO_ESTACION

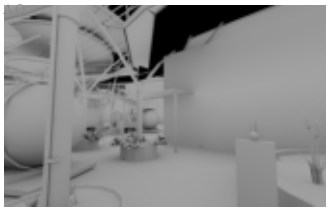
Atrio del Metro

marquesina volada y bolardos

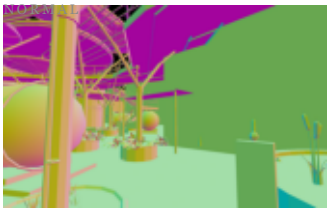


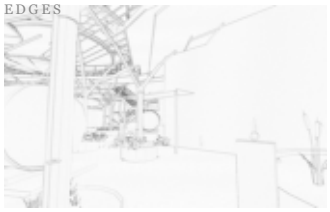


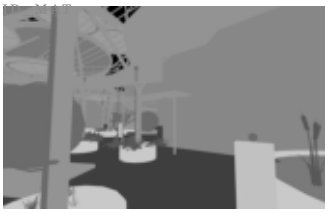












CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
F3_ATRIO_ESTACION	50.0 6.0 1.6	57.0 -2.0 2.5	65.0°	19 mm	0.10–1200 m

PACK · F4_CARLOS_III

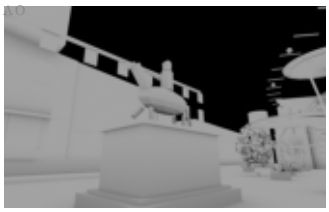
Carlos III

la estatua con el refugio detrás

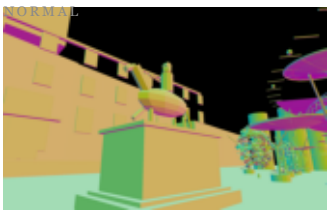














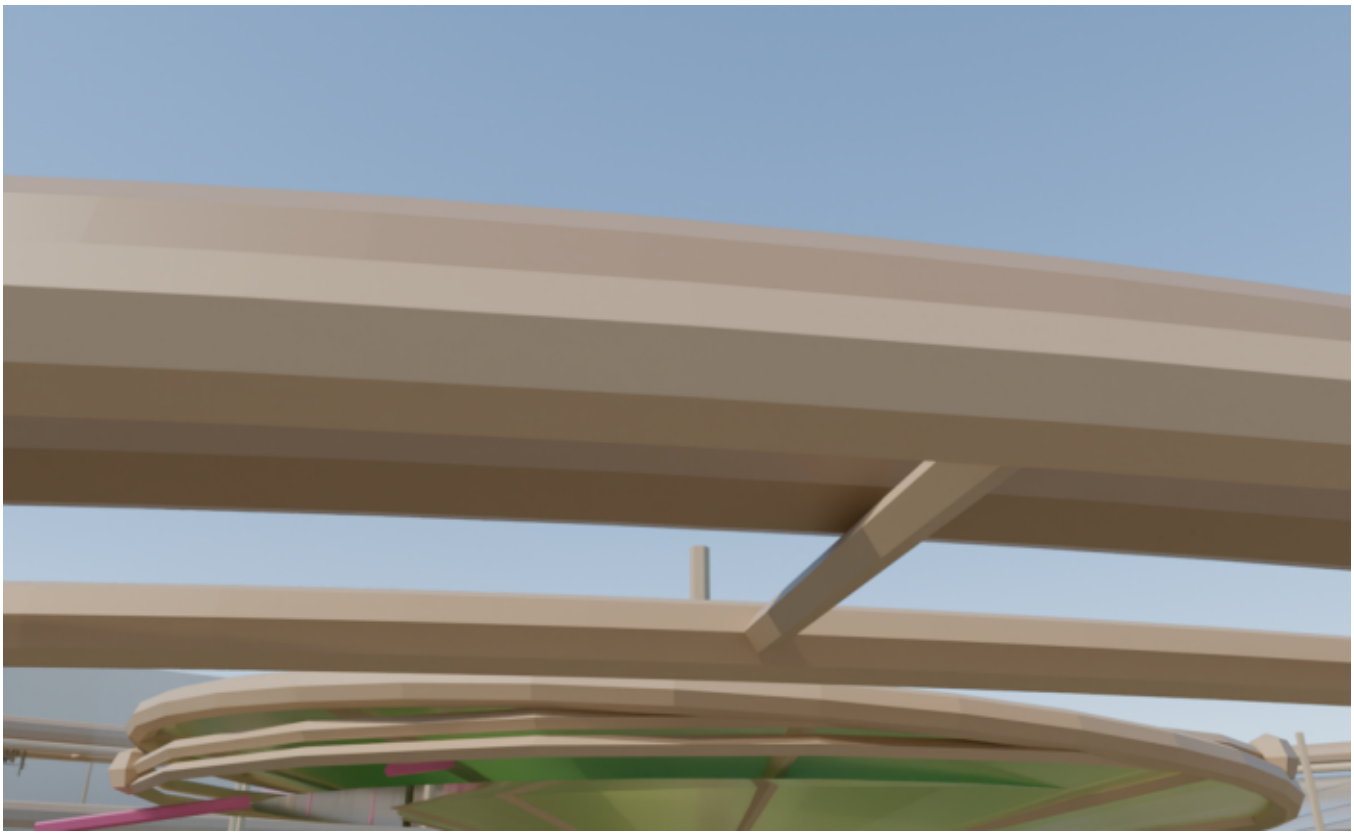


CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
F4_CARLOS_III	-22.0 -8.0 1.6	-28.0 -14.0 4.0	55.0°	23 mm	0.10–1200 m

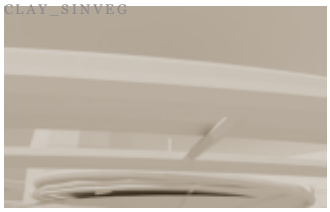
PACK · F5_DETALLE_NUDO

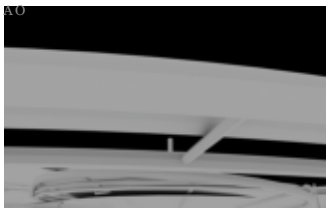
Nudo mástil-hoja

cables, casquillos y tensores 1:5

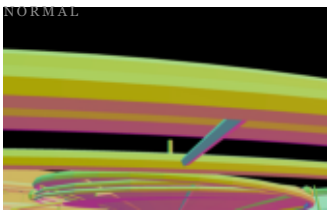


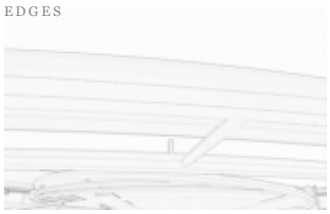


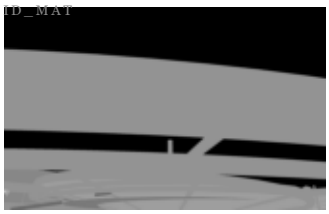












CÁMARA	POSICIÓN (X Y Z)	OBJETIVO	FOV V	FOCAL EQ.	CLIP
F5_DETALLE_NUDO	-1.6 3.2 8.1	0.1 5.1 9.2	42.0°	31 mm	0.10–1200 m